



# AF 745 Cationic Polymer (AFCO 18024)

## Safety Data Sheet

according to Federal Register / Vol. 77, No. 58 / Monday, March 26, 2012 / Rules and Regulations  
Revision Date: 02/19/2024

Version: 1.0

### SECTION 1: IDENTIFICATION

#### Product Identifier

**Product Form:** Mixture

**Product Name:** AF 745 Cationic Polymer

**Product Code:** AFCO 18024

#### Intended Use of the Product

**Use of the Substance/Mixture:** Wastewater processing aid for industrial applications. For professional use only.

#### Name, Address, and Telephone of the Responsible Party

##### **Company**

AFCO  
800 Development Avenue  
Chambersburg, PA 17201  
T: 800-345-1329  
[www.afcocare.com](http://www.afcocare.com)

#### Emergency Telephone Number

**Emergency Number** : 1-800-424-9300 (CHEMTREC)

### SECTION 2: HAZARDS IDENTIFICATION

#### Classification of the Substance or Mixture

Classification according to paragraph (d) of 29 CFR 1910.1200:

**Classification (GHS-US)** : Not classified.

#### Label Elements

Labelling according to paragraph (f) of 29 CFR 1910.1200:

**GHS-US Labeling** : No labelling required.

**Signal Word (GHS-US)** : None.

**Hazard Statements (GHS-US)** : None.

**Precautionary Statements (GHS-US)** : None.

#### Other Hazards

**Other Hazards:** Aqueous solutions or powders that become wet render surfaces extremely slippery.

For explanation of abbreviations see Section 16.

**Unknown Acute Toxicity (GHS-US)** Not available.

### SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

**Substances:** Not applicable, this product is a mixture.

#### Mixture

| Name of Hazardous Ingredients | Product identifier | % (w/w)  | Classification (GHS-US)                    |
|-------------------------------|--------------------|----------|--------------------------------------------|
| Sulfamic acid                 | (CAS No) 5329-14-6 | 2.5 - 10 | Skin Irrit. 2, H315<br>Eye Irrit. 2A, H319 |

Full text of H-phrases: see section 16.

### SECTION 4: FIRST AID MEASURES

#### Description of First Aid Measures

**General:** Never give anything by mouth to an unconscious person. If you feel unwell, seek medical advice (show the label where possible).

**Inhalation:** Move to fresh air. Get medical attention if symptoms occur.

**Skin Contact:** Wash off with soap and plenty of water. Get medical attention if irritation develops and persists.

**Eye Contact:** Rinse immediately with plenty of water, also under the eyelids. Get medical attention.

**Ingestion:** Rinse mouth. If conscious, give the victim plenty of water to drink. Induce vomiting, but only if victim is fully conscious.

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### **Most Important Symptoms and Effects Both Acute and Delayed**

Powder can cause localized skin irritation in folds of the skin or under tight clothing. Contact with dust can cause mechanical irritation or drying of the skin.

### **Indication of Any Immediate Medical Attention and Special Treatment Needed**

None reasonably foreseeable.

## **SECTION 5: FIRE-FIGHTING MEASURES**

### **Extinguishing Media**

**Suitable Extinguishing Media:** Water. Water spray. Foam. Carbon dioxide (CO<sub>2</sub>). Dry powder.

Warning! Aqueous solutions or powders that become wet render surfaces extremely slippery.

**Unsuitable Extinguishing Media:** None known.

### **Special Hazards Arising From the Substance or Mixture**

**Fire Hazard:** Not flammable.

**Explosion Hazard:** Product is not explosive.

**Reactivity:** Spilled product is slippery underfoot. Very slippery when wet.

### **Advice for Firefighters**

**Precautionary Measures Fire:** Exercise caution when fighting any chemical fire.

**Firefighting Instructions:** Use water spray or fog for cooling exposed containers.

**Protection During Firefighting:** Do not enter fire area without proper protective equipment. Wear self contained breathing apparatus for firefighting if necessary.

**Hazardous Combustion Products:** Thermal decomposition may produce: Carbon oxides (CO, CO<sub>2</sub>). Nitrogen Oxides (NO<sub>x</sub>). Hydrogen chloride gas. Hydrogen cyanide (hydrocyanic acid) may be produced in the event of combustion in an oxygen deficient atmosphere.

**Other information:** Aqueous solutions or powders that become wet render surfaces extremely slippery.

### **Other Information**

Aqueous solutions or powders that become wet render surfaces extremely slippery.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **Personal Precautions, Protective Equipment and Emergency Procedures**

**General Measures:** Avoid contact with skin and eyes. Avoid dust formation. Avoid breathing dust. Aqueous solutions or powders that become wet render surfaces extremely slippery.

#### **For Non-Emergency Personnel**

**Protective Equipment:** Wear adequate personal protective equipment (PPE). (see Section 8 Exposure Controls/Personal Protection).

**Emergency Procedures:** Keep people away from spill/leak. Prevent further leakage or spillage if safe to do so.

#### **For Emergency Personnel**

**Protective Equipment:** Wear adequate personal protective equipment (PPE). (see Section 8 Exposure Controls/Personal Protection).

**Emergency Procedures:** Keep people away from spill/leak. Prevent further leakage or spillage if safe to do so.

### **Environmental Precautions**

As with all chemical products, do not flush into surface water.

### **Methods and Material for Containment and Cleaning Up**

**Small Spills:** Do not flush with water. Clean up promptly by sweeping or vacuum.

**Large Spills:** Do not flush with water. Prevent unauthorized access. Sweep up and shovel into suitable containers for disposal.

**Residues:** After cleaning, flush with traces of water.

### **Reference to Other Sections**

SECTION 7: Handling and storage; SECTION 8: Exposure controls/personal protection; SECTION 13: Disposal considerations.

## **SECTION 7: HANDLING AND STORAGE**

### **Precautions for Safe Handling**

Avoid contact with skin and eyes. Avoid dust formation. Avoid breathing dust. Wash hands before breaks and at the end of workday.

### **Conditions for Safe Storage, Including Any Incompatibilities**

**Technical Measures:** Comply with applicable regulations.

**Storage Conditions:** Keep in a dry place.

**Incompatible Materials:** Oxidizing agents.

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**Specific End Use(s):** Wastewater processing aid for industrial applications. For professional use only.

### SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Control Parameters

None known.

#### Exposure Controls

**Appropriate Engineering Controls:** Emergency eye wash fountains and safety showers should be available in the immediate vicinity of any potential exposure. Ensure all national/local regulations are observed. Ensure adequate ventilation, especially in confined areas. Use local exhaust if dusting occurs. Natural ventilation is adequate in absence of dusts.

**Personal Protective Equipment:** Protective goggles. Protective clothing. Gloves.



**Materials for Protective Clothing:** Chemically resistant materials and fabrics.

**Hand Protection:** Wear PVC or other plastic material gloves.

**Eye Protection:** Safety glasses with side-shields.

**Skin and Body Protection:** Workclothes protecting arms, legs and body.

**Respiratory Protection:** No personal respiratory protective equipment normally required. Dust safety masks recommended where working powder concentration is more than 10 mg/m<sup>3</sup>.

**Additional Information:** Wash hands before breaks and at the end of workday. Handle in accordance with good industrial hygiene and safety practice.

**Environmental Exposure Controls:** Do not allow uncontrolled discharge of product into the environment. Do not flush into surface water.

### SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

#### Information on Basic Physical and Chemical Properties

|                                                    |                                                                 |
|----------------------------------------------------|-----------------------------------------------------------------|
| Physical State                                     | : Solid.                                                        |
| Appearance                                         | : White, granular.                                              |
| Odor                                               | : None.                                                         |
| Odor Threshold                                     | : Not applicable.                                               |
| pH                                                 | : 2.5 - 4.5 @ 5 g/L                                             |
| Relative Evaporation Rate (butylacetate=1)         | : Not available.                                                |
| Melting Point                                      | : >100°C (212°F)                                                |
| Freezing Point                                     | : >100°C (212°F)                                                |
| Boiling Point                                      | : Not applicable.                                               |
| Flash Point                                        | : Not applicable.                                               |
| Auto-ignition Temperature                          | : Not applicable.                                               |
| Decomposition Temperature                          | : >200°C (392°F)                                                |
| Flammability (solid, gas)                          | : Not combustible.                                              |
| Lower Flammable Limit                              | : Not expected to create explosive atmospheres.                 |
| Upper Flammable Limit                              | : Not expected to create explosive atmospheres.                 |
| Vapor Pressure                                     | : Not applicable.                                               |
| Relative Density                                   | : 0.6 - 0.9                                                     |
| Solubility                                         | : Soluble in water.                                             |
| Partition coefficient: n-octanol/water (log value) | : <0                                                            |
| Viscosity                                          | : Not available.                                                |
| Kinematic Viscosity                                | : No data available.                                            |
| Explosive Properties                               | : Not expected to be explosive based on the chemical structure. |
| Oxidizing Properties                               | : Not expected to be oxidizing based on the chemical structure. |
| Particle Characteristics                           | : No data available.                                            |

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### SECTION 10: STABILITY AND REACTIVITY

**Reactivity:** Hazardous polymerization does not occur.

**Chemical Stability:** Stable.

**Possibility of Hazardous Reactions:** Oxidizing agents may cause exothermic reactions.

**Conditions to Avoid:** None known.

**Incompatible Materials:** Oxidizing agents.

**Hazardous Decomposition Products:** Thermal decomposition may produce: Carbon oxides (CO, CO<sub>2</sub>). Nitrogen Oxides (NO<sub>x</sub>).

Hydrogen chloride gas. Hydrogen cyanide (hydrocyanic acid) may be produced in the event of combustion in an oxygen deficient atmosphere.

### SECTION 11: TOXICOLOGICAL INFORMATION

#### Information on Toxicological Effects – Product:

##### Acute Toxicity:

| AF 745 Cationic Polymer (AFCO 18024) |                                                        |
|--------------------------------------|--------------------------------------------------------|
| LD50 Oral Rat                        | > 5000 mg/kg                                           |
| LD50 Dermal Rat                      | > 5000 mg/kg                                           |
| LC50 Inhalation                      | The product is not expected to be toxic by inhalation. |

**Skin Corrosion/Irritation:** Not irritating.

**Serious Eye Damage/Irritation:** Testing conducted according to the Draize technique showed the material produces no corneal or iridial effects and only slight transitory conjunctival effects similar to those which all granular materials have on conjunctiva.

**Respiratory or Skin Sensitization:** The results of testing on guinea pigs showed this material to be non-sensitizing.

**Mutagenicity:** Not mutagenic.

**Teratogenicity:** Not available.

**Carcinogenicity:** Not carcinogenic.

**Specific Target Organ Toxicity (Repeated Exposure):** No known effects.

**Reproductive Toxicity:** Not toxic for reproduction.

**Specific Target Organ Toxicity (Single Exposure):** No known effects.

**Aspiration Hazard:** No hazards resulting from the material as supplied.

#### Information on Acute Toxicity of Hazardous – Components:

| Sulfamic acid (5329-14-6) |                                                        |
|---------------------------|--------------------------------------------------------|
| LD50 Oral Rat             | = 2065 - 2140 mg/kg                                    |
| NOAEL/ Dermal Rat         | = 2000 mg/kg (OECD 402)                                |
| Inhalation                | The product is not expected to be toxic by inhalation. |

**Skin Corrosion/Irritation:** Not irritating. (OECD 404) (SNF).

**Serious Eye Damage/Irritation:** Moderately irritating to the eyes. (EPA OPPTS 870.2400).

**Respiratory or Skin Sensitization:** The product is not expected to be sensitizing.

**Mutagenicity:** Negative in the Ames Test (OECD 471). Negative in the In vitro Mammalian Cell Gene Mutation Test (OECD 476). Not mutagenic. (OECD 472, 487).

**Teratogenicity:** Not available.

**Carcinogenicity:** Based on the absence of mutagenicity, it is unlikely that the substance is carcinogenic.

**Specific Target Organ Toxicity (Repeated Exposure):** No known effects.

**Reproductive Toxicity:** Based on available data, product is not expected to be toxic for reproduction.

Prenatal Development Toxicity Study (OECD 414).

- NOAEL/Maternal toxicity/rat = 200 mg/kg/day.

- NOAEL/Developmental toxicity/rat = 200 mg/kg/day.

**Specific Target Organ Toxicity (Single Exposure):** No known effects.

**Aspiration Hazard:** No known effects.

### SECTION 12: ECOLOGICAL INFORMATION

#### Toxicity

| AF 745 Cationic Polymer (AFCO 18024) |                                                                      |
|--------------------------------------|----------------------------------------------------------------------|
| LC50 Fish                            | 5-10 mg/l (Exposure time: 96 h - Species: Danio rerio) (OECD 203)    |
| EC50 Invertebrates                   | 20-50 mg/l (Exposure time: 48 h - Species: Daphnia magna) (OECD 202) |

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|                                   |                                                                                                                                                                                                |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Algae                             | Algal inhibition tests are not appropriate. The flocculation characteristics of this product interfere directly in the test medium preventing homogeneous distribution which invalidates test. |
| Chronic Toxicity to Fish          | No data available.                                                                                                                                                                             |
| Chronic Toxicity to Invertebrates | No data available.                                                                                                                                                                             |
| Toxicity to Microorganisms        | No data available.                                                                                                                                                                             |
| Effects on Terrestrial Organisms  | No data available. Readily biodegradable, exposure to soil is unlikely.                                                                                                                        |
| Sediment Toxicity                 | No data available. Readily biodegradable, exposure to sediment is unlikely.                                                                                                                    |
| <b>Sulfamic acid (5329-14-6)</b>  |                                                                                                                                                                                                |
| LC50 Fish                         | 70.3 mg/l (Exposure time: 96 h - Species: Pimephales promelas) (OECD 203)                                                                                                                      |
| EC50 Invertebrates                | 71.6 mg/l (Exposure time: 48 h - Species: Daphnia magna) (OECD 202)                                                                                                                            |
| IC50 Algae                        | 48 mg/l (Exposure time: 72 h - Species: Selenastrum capricornutum) (OECD 201)                                                                                                                  |
| Chronic Toxicity to Fish          | NOECD/Danio rerio/34 days >= 60 mg/l (OECD 210)                                                                                                                                                |
| Chronic Toxicity to Invertebrates | NOECD/Daphnia magna/21 days = 19 mg/l (OECD 211)                                                                                                                                               |
| Toxicity to Microorganisms        | EC50/activated sludge/3 hrs >200 mg/l (OECD 209)                                                                                                                                               |
| Effects on Terrestrial Organisms  | No data available.                                                                                                                                                                             |
| Sediment toxicity                 | No data available.                                                                                                                                                                             |

### Persistence and Degradability

|                                             |                                                                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AF 745 Cationic Polymer (AFCO 18024)</b> |                                                                                                                                                       |
| Degradability                               | Based on the degradability data of the components, this product is expected to be readily (bio)degradable according to OECD criteria.                 |
| Hydrolysis                                  | At natural pHs (>6) the polymer degrades due to hydrolysis to more than 70% in 28 days. The hydrolysis products are not harmful to aquatic organisms. |
| Photolysis                                  | No data available.                                                                                                                                    |
| <b>Sulfamic acid (5329-14-6)</b>            |                                                                                                                                                       |
| Degradation                                 | Not relevant. (inorganic).                                                                                                                            |
| Hydrolysis                                  | Does not hydrolyze.                                                                                                                                   |
| Photolysis                                  | No data available.                                                                                                                                    |

### Bioaccumulative Potential

|                                             |                                |
|---------------------------------------------|--------------------------------|
| <b>AF 745 Cationic Polymer (AFCO 18024)</b> |                                |
| Bioaccumulative Potential                   | Not expected to bioaccumulate. |
| <b>Sulfamic acid (5329-14-6)</b>            |                                |
| Partition coefficient (Log Pow)             | -4.34 @ 20°C                   |
| Bioconcentration factor BCF                 | ~0                             |

### Mobility in Soil

No data available for product or components.

### Other Adverse Effects

None known.

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste Disposal Recommendations:** Dispose in accordance with local and national regulations. Can be landfilled or incinerated, when in compliance with local regulations.

**Waste Packaging:** Rinse empty containers with water and use the rinse-water to prepare the working solution. If recycling is not practicable, dispose of in compliance with local regulations. Can be landfilled or incinerated, when in compliance with local regulations.

## SECTION 14: TRANSPORT INFORMATION

|                                     |   |                              |
|-------------------------------------|---|------------------------------|
| <b>14.1 In Accordance with DOT</b>  | : | Not regulated for transport. |
| <b>14.2 In Accordance with IMDG</b> | : | Not regulated for transport. |
| <b>14.3 In Accordance with IATA</b> | : | Not regulated for transport. |
| <b>14.4 In Accordance with TDG</b>  | : | Not regulated for transport. |

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### SECTION 15: REGULATORY INFORMATION

|                                                                                                                                  |                     |
|----------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>US Federal Regulations</b>                                                                                                    |                     |
| <b>AF 745 Cationic Polymer (AFCO 18024)</b>                                                                                      |                     |
| All components are either listed or exempt from being listed on the United States TSCA (Toxic Substances Control Act) inventory. |                     |
| <b>SARA Section 311/312 Hazard Class</b>                                                                                         | Not concerned.      |
| <b>SARA Title III Sections</b>                                                                                                   |                     |
| <b>SARA 302 (TPQ) - Reportable Quantity</b>                                                                                      | Not concerned.      |
| <b>SARA 304 - Reportable Quantity</b>                                                                                            | Not concerned.      |
| <b>SARA 313 (De Minimis Concentration)</b>                                                                                       | Not concerned.      |
| <b>Clean Water Act Section 311 Hazardous Substances (40 CFR 117.3) - RQ</b>                                                      | Not concerned.      |
| <b>Clean Air Act (CAA) Section 112(r) Accidental Release (40 CFR 68) - RQ</b>                                                    | Not concerned.      |
| <b>CERCLA Hazardous Substance List (40 CFR 302.4) - RQ</b>                                                                       | Not concerned.      |
| <b>RCRA Status</b>                                                                                                               | Not RCRA hazardous. |

|                                                                                                                                                      |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>US State Regulations</b>                                                                                                                          |  |
| <b>U.S. - California - Proposition 65</b>                                                                                                            |  |
| WARNING! This product contains a chemical known to the State of California to cause cancer and birth defects or other reproductive harm, Acrylamide. |  |

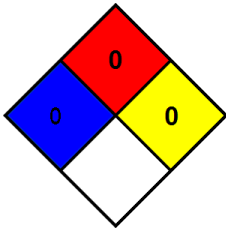
|                                                                                                                                                                                   |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Canadian Regulations</b>                                                                                                                                                       |  |
| <b>Sulfamic acid (5329-14-6)</b>                                                                                                                                                  |  |
| Listed on the Canadian DSL (Domestic Substances List) inventory.                                                                                                                  |  |
| This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the SDS contains all of the information required by CPR. |  |

### SECTION 16: OTHER INFORMATION, INCLUDING DATE OF PREPARATION OR LAST REVISION

|                          |   |                                                                                                                                     |
|--------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Revision date</b>     | : | 02/17/2024                                                                                                                          |
| <b>Other Information</b> | : | This document has been prepared in accordance with the SDS requirements of the OSHA Hazard Communication Standard 29 CFR 1910.1200. |

|                               |                                                |
|-------------------------------|------------------------------------------------|
| <b>GHS Full Text Phrases:</b> |                                                |
| Eye Irrit. 2A                 | Serious eye damage/eye irritation Category 2A. |
| Skin Irrit. 2                 | Skin corrosion/irritation Category 2.          |
| H315                          | Causes skin irritation.                        |
| H319                          | Causes serious eye irritation.                 |

|                           |   |                                                                                                         |
|---------------------------|---|---------------------------------------------------------------------------------------------------------|
| <b>NFPA Health Hazard</b> | : | 0 - Exposure under fire conditions would offer no hazard beyond that of ordinary combustible materials. |
| <b>NFPA Fire Hazard</b>   | : | 0 - Materials that will not burn.                                                                       |
| <b>NFPA Reactivity</b>    | : | 0 - Normally stable, even under fire exposure conditions, and are not reactive with water.              |



|                     |                                                       |
|---------------------|-------------------------------------------------------|
| <b>HMIS Rating</b>  |                                                       |
| <b>Health</b>       | : 0 - Minimal Hazard - No significant risk to health. |
| <b>Flammability</b> | : 0 - Minimal Hazard.                                 |
| <b>Physical</b>     | : 0 - Minimal Hazard.                                 |

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

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### Party Responsible for the Preparation of This Document

AFCO  
800 Development Avenue  
Chambersburg, PA 17201  
T: 800-345-1329

*This information is based on our current knowledge and is intended to describe the product for the purposes of health, safety and environmental requirements only. It should not therefore be construed as guaranteeing any specific property of the product.*

North America GHS SDS 2015 (U.S., Can., Mex.)

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